AnthracoForm

The anthracological protocol of the laboratory and how it is implemented in AnthracoForm

Supplementary Material S3 (summary). User guide: S2. Data dictionary: S1.

1. Purpose of this supplement

AnthracoForm records charcoal identified fragment by fragment, whether the material comes from a full sampling protocol or reaches the laboratory without a sampling design, for example charcoal collected by hand during a rescue excavation. The record is always the same; only what can be calculated from it changes. This supplement does not prescribe how to sample. It describes how our laboratory (CASEs, Universitat Pompeu Fabra) samples and analyses charcoal, because AnthracoForm was developed together with this protocol, and shows where each part of the protocol is implemented in the form.

2. Recovery channels

The origin of a fragment is not an administrative detail: it determines what the fragment can tell. All material is recorded with its channel, and channels are never added together.

Channel	How it is collected	What it is used for
HF, LF	Heavy and light fractions of the same sample of known volume (VOL; usually 20 or 40 L), separated by flotation	Richness, abundances, densities, saturation curves
DS	Dry sieving in the field, 4 mm mesh; sediment volume usually not recorded	Richness only (conditional discovery curve)
HC	Hand collection during excavation, picked by eye	Richness only; operator training and photographic reference collection
STR	Closed context: hearth, oven, dump, concentration of charcoal distinct from the matrix	Analysis by event, separate from the matrix
AC	Contents of an artefact (lamp, amphora, vessel), emptied before washing	Analysis by object, separate from the matrix
C14	Fragments selected for dating	Outside the quantitative workflow; identified before being sent

Table S3.1. *Recovery channels of the protocol.*

- **Richness yes, abundance no.** Every identified fragment, from any channel, enters the taxa list of the unit and the site ubiquity. Only HF and LF, which share a known volume, enter abundances, percentages, densities and saturation curves: fragments collected by hand during excavation are not comparable with those recovered by flotation (Smart and Hoffman 1988, 173–176).
- **DS is never added to HF and LF.** Even when the total volume of dry-sieved sediment is recorded, recovery is not comparable with flotation: dry sieving detects the presence of taxa (often rare or dense taxa that flotation may miss), not their quantity.

3. Unit of analysis

The minimum unit is the SU together with its context, not the SU alone: the context can subdivide an SU or mark a closed context within it, and merging the contexts of an SU destroys an association

that cannot be recovered. The ordinary dispersed matrix is the VOL context. A unit that, once sorting is complete, yields neither identifiable charcoal nor other macroremains is recorded as sterile: this is a result, not a reason to discard the unit.

4. Workflow

The phases are carried out in order, and no SU moves to the next phase before the previous one is closed.

- **Preliminary sorting (site).** All HC, DS, STR and AC material is sieved through the laboratory column (4, 2, 1 and 0.5 mm; HC is not sieved) and cleaned: seeds, twigs, bones and any other non-charcoal material are separated, so that the following phases deal only with charcoal. Candidates for radiocarbon dating are sent immediately.
- **Phase 0 — training and reference collection (HC, site).** Before any SU is analysed, up to 20 HC fragments per SU are studied, with SUs taken in random order: the largest among those at least 6 mm long, not friable, little or not vitrified and free of concretions (DS of the same quality if HC is missing). Each is photographed in the three sections and recorded. The site discovery curve (TDC) shows when the operator has seen enough local woods to recognise them: it is an internal check, not a saturation curve or a result.
- **Phase 1 — sample of known volume (HF and LF, by SU).** The fractions are dry-sieved into > 4, 4–2, 2–1, 1–0.5 and < 0.5 mm. Each whole LF fraction is weighed and measured before anything is removed; HF is not, because stones are discarded in the field. Non-charcoal material is separated, and charcoal ≥ 2 mm is weighed. Fragments are identified from the four cells HF > 4, HF 4–2, LF > 4 and LF 4–2 in proportion to their abundance, always taking from the cell furthest behind, so that no cell is exhausted before the others. Cells with more than about 150 fragments are sub-sampled on a grid (10% of the weight), and counts are rescaled cell by cell before being added together.
- **Phase 2 — dry sieving (DS, by SU).** The conditional discovery curve (CDC) starts from the taxa list of HF and LF of the same unit and counts only the taxa added by DS: a flat curve is a result.
- **Phase 3 — closed contexts (STR and AC, by event or by object).** Percentages within the context are allowed, but they are never added to the dispersed matrix of the SU: the context is read against it.
- **Phase 4 — radiocarbon (C14).** Large, well-preserved, short-lived fragments are chosen (twigs, seeds, wood with few rings), avoiding long-lived woods (old wood effect). They keep the channel from which they were separated and are not treated with deflocculant or recycled water.

5. When to stop

In our laboratory the identification of an SU is closed when the cumulative HF+LF curve stays flat for 50 consecutive fragments without new taxa and each cell has at least 8 identified fragments (or is exhausted); the TDC and the CDC are closed in the same way. This is a working rule, not a statistical test: the plateau of a curve has no mathematically defined end point (Kabukcu and Chabal 2021). For this reason AnthracoForm never stops the recording: it shows the curve and the thresholds of 100, 200, 250, 300 and 400 fragments, and the decision remains with the analyst.

6. Sites with a single channel

When only LF is available, Phase 1 works on two cells (LF > 4 and LF 4–2), identification is limited to fragments ≥ 2 mm, and abundances and densities refer to the light fraction only, which is filtered by density: dense woods may be under-represented. This must be stated in the results. Phases without material are skipped, and their absence is documented.

7. From the protocol to AnthracoForm

In the protocol	In AnthracoForm
Recovery channel	Field channel (HF, LF, DS, HC, STR, C14). Contents of artefacts (AC) are recorded as STR, with the object in the context field
VOL sample, bag	id_samp; an empty context is the VOL matrix
Unit SU + context	su and context: unit of the curves and of the summary
Order of identification	id_det, assigned automatically: curves follow the actual order
Size classes	fraction (gt4, 4to2, 2to1, 1to05)
HF+LF curves, CDC, TDC	Panel always visible next to the identification, with the curve drawn; curve windows and SU summary (S2, Section 7)
Anatomical features	IAWA features by section: 0 absent, 1 present, 2 required present (present and diagnostic), 3 required absent (absent and diagnostic), empty not assessed, as the a, p, r, e codes of InsideWood (S2, Table 2); observed sections
Taphonomy	Eight alterations (1 or empty) and other_alteration as free text
Phase 0: photographs of the three sections	Photographs by section in the project photo folder; taxon report with all its photographs (S2, Section 8)
Woods not yet identified	Provisional types (tipo, tipo_desc, tipo_resolved), which can be compared with the taxa and have their own report
Weights, volumes, sub-sampling, sterility, samples sent for C14	Outside the form: separate tables, one per unit of observation, joined to the AnthracoForm file by SU and context

Table S3.2. *Where each part of the protocol is implemented in AnthracoForm.*

References

Kabukcu, C. and Chabal, L. 2021. Sampling and quantitative analysis methods in anthracology from archaeological contexts: achievements and prospects. *Quaternary International* 593–594: 6–18. https://doi.org/10.1016/j.quaint.2020.11.004

Smart, T.L. and Hoffman, E.S. 1988. Environmental interpretation of archaeological charcoal. In: Hastorf, C.A. and Popper, V.S. (eds.) *Current Paleoethnobotany: Analytical Methods and Cultural Interpretations of Archaeological Plant Remains*. Chicago: University of Chicago Press, 167–205.